

## ZIELONA GORA, Poland

WMO: 124000

Lat: 51.9299N

Lon: 15.5247E

Elev: 192

StdP: 99.03

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.7      | -9.6 | -15.7                         | 1.0 | -11.4 | -12.7 | 1.3 | -7.8 | 9.3                   | 6.0  | 8.4 | 5.5  | 2.7                   | 70   | 0.508 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.7                    | 30.1            | 19.2 | 28.2 | 18.5 | 26.4 | 17.9 | 20.6                | 27.2 | 19.7 | 25.6 | 18.9 | 24.4 | 3.0                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.6                            | 13.7 | 22.4 | 17.7 | 13.0 | 21.6 | 16.9 | 12.3 | 21.2 | 60.3          | 27.5 | 57.2 | 25.4 | 54.2 | 24.5 | 24.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 7.4           | 6.5  | 5.7 | DB                         | -14.8 | 33.4               | 4.1 | 1.6                                                | -17.7 | 34.6       | -20.1 | 35.6       | -22.4 | 36.5       | -25.3 | 37.7 | (4) |
| (5)               |      |     | WB                         | -15.1 | 22.4               | 4.0 | 1.0                                                | -18.0 | 23.1       | -20.3 | 23.6       | -22.6 | 24.2       | -25.5 | 24.9 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.5   | -0.7  | 0.7   | 4.0   | 9.5   | 14.0  | 17.4  | 19.6  | 19.2  | 14.6  | 9.6   | 4.5   | 0.8  | (6)  |
| (7)  |                                                                                  | DBStd       | 8.28  | 5.08  | 4.78  | 4.37  | 4.50  | 3.99  | 3.73  | 3.54  | 3.31  | 3.35  | 3.99  | 3.93  | 4.50 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1368  | 331   | 260   | 190   | 62    | 9     | 0     | 0     | 2     | 58    | 169   | 287   |      | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3403  | 589   | 493   | 443   | 265   | 143   | 61    | 27    | 29    | 119   | 272   | 417   | 544  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1182  | 0     | 1     | 4     | 48    | 134   | 223   | 298   | 286   | 141   | 44    | 3     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 177   | 0     | 0     | 0     | 2     | 10    | 34    | 67    | 56    | 8     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 1340  | 0     | 0     | 0     | 14    | 82    | 266   | 528   | 394   | 55    | 1     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 356   | 0     | 0     | 0     | 1     | 12    | 71    | 161   | 102   | 9     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.0   | 3.4   | 3.4   | 3.3   | 3.1   | 2.8   | 2.8   | 2.7   | 2.6   | 2.7   | 2.8   | 3.0   | 3.4  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 601   | 44    | 35    | 45    | 33    | 54    | 60    | 90    | 67    | 48    | 39    | 44    | 41   | (15) |
| (16) |                                                                                  | PrecMax     | 772   | 105   | 87    | 126   | 70    | 128   | 115   | 208   | 131   | 142   | 100   | 120   | 100  | (16) |
| (17) |                                                                                  | PrecMin     | 397   | 2     | 5     | 17    | 2     | 10    | 12    | 20    | 17    | 6     | 9     | 1     | 13   | (17) |
| (18) |                                                                                  | PrecStd     | 106   | 22    | 19    | 25    | 18    | 29    | 27    | 52    | 34    | 29    | 23    | 24    | 19   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 11.4  | 13.1  | 18.0  | 25.3  | 28.3  | 31.6  | 32.8  | 32.4  | 28.0  | 22.1  | 14.9  | 11.5 | (19) |
| (20) |                                                                                  |             | MCWB  | 8.6   | 8.8   | 10.5  | 15.2  | 17.6  | 20.1  | 19.7  | 20.0  | 18.5  | 16.2  | 11.9  | 9.4  | (20) |
| (21) |                                                                                  | 2%          | DB    | 9.1   | 10.5  | 14.7  | 22.1  | 25.7  | 28.7  | 30.4  | 29.3  | 24.8  | 19.2  | 12.8  | 9.7  | (21) |
| (22) |                                                                                  |             | MCWB  | 7.4   | 7.6   | 9.0   | 13.5  | 16.6  | 18.7  | 19.1  | 19.3  | 16.9  | 14.6  | 10.4  | 8.0  | (22) |
| (23) |                                                                                  | 5%          | DB    | 7.4   | 8.8   | 12.3  | 19.2  | 23.4  | 26.3  | 28.3  | 27.1  | 22.6  | 17.1  | 11.2  | 8.1  | (23) |
| (24) |                                                                                  |             | MCWB  | 5.7   | 6.3   | 8.0   | 12.0  | 15.5  | 17.5  | 18.5  | 18.5  | 16.1  | 13.6  | 9.4   | 6.7  | (24) |
| (25) |                                                                                  | 10%         | DB    | 5.8   | 7.2   | 10.3  | 17.0  | 21.2  | 24.2  | 26.3  | 25.4  | 20.4  | 15.2  | 9.9   | 6.6  | (25) |
| (26) |                                                                                  |             | MCWB  | 4.5   | 5.2   | 6.9   | 10.8  | 14.2  | 16.6  | 18.0  | 17.9  | 15.2  | 12.7  | 8.4   | 5.5  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 9.4   | 9.7   | 11.6  | 16.2  | 18.7  | 21.7  | 21.6  | 21.6  | 19.4  | 16.8  | 12.6  | 9.9  | (27) |
| (28) |                                                                                  |             | MCDWB | 10.4  | 11.6  | 16.2  | 23.4  | 25.6  | 29.2  | 28.3  | 28.8  | 25.2  | 21.1  | 14.3  | 11.0 | (28) |
| (29) |                                                                                  | 2%          | WB    | 7.6   | 8.1   | 9.9   | 14.1  | 17.4  | 19.8  | 20.6  | 20.5  | 18.0  | 15.2  | 10.8  | 8.2  | (29) |
| (30) |                                                                                  |             | MCDWB | 9.0   | 10.0  | 13.5  | 20.4  | 23.8  | 26.1  | 27.0  | 26.8  | 22.8  | 18.3  | 12.1  | 9.4  | (30) |
| (31) |                                                                                  | 5%          | WB    | 5.9   | 6.7   | 8.6   | 12.6  | 16.3  | 18.6  | 19.7  | 19.6  | 16.9  | 14.0  | 9.7   | 6.8  | (31) |
| (32) |                                                                                  |             | MCDWB | 7.2   | 8.5   | 11.5  | 18.1  | 21.9  | 24.4  | 25.7  | 25.1  | 21.1  | 16.2  | 11.0  | 7.9  | (32) |
| (33) |                                                                                  | 10%         | WB    | 4.6   | 5.3   | 7.4   | 11.4  | 15.1  | 17.4  | 18.8  | 18.7  | 15.9  | 12.9  | 8.5   | 5.5  | (33) |
| (34) |                                                                                  |             | MCDWB | 5.7   | 7.1   | 9.9   | 16.2  | 19.8  | 22.5  | 24.4  | 23.6  | 19.5  | 15.0  | 9.7   | 6.5  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 4.4   | 5.3   | 6.6   | 8.6   | 9.0   | 8.9   | 8.7   | 8.8   | 7.8   | 6.4   | 4.8   | 4.1  | (35) |
| (36) |                                                                                  |             | MCDBR | 5.7   | 7.1   | 9.9   | 12.3  | 12.4  | 12.2  | 11.7  | 11.7  | 11.3  | 8.8   | 6.7   | 5.4  | (36) |
| (37) |                                                                                  |             | MCWBR | 4.6   | 4.9   | 5.7   | 6.0   | 5.7   | 5.2   | 4.2   | 4.8   | 5.3   | 5.1   | 4.8   | 4.4  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 5.8   | 6.4   | 8.7   | 11.1  | 10.9  | 11.2  | 10.3  | 10.3  | 9.9   | 8.1   | 6.5   | 5.2  | (38) |
| (39) |                                                                                  |             | MCWBR | 4.9   | 5.0   | 5.7   | 5.7   | 5.4   | 5.4   | 4.4   | 4.8   | 5.3   | 5.0   | 5.0   | 4.4  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.312 | 0.336 | 0.373 | 0.413 | 0.409 | 0.407 | 0.427 | 0.422 | 0.382 | 0.360 | 0.338 | 0.306 | (40) |      |
| (41) |                                                                                  | taud        | 2.355 | 2.341 | 2.263 | 2.209 | 2.266 | 2.299 | 2.251 | 2.281 | 2.366 | 2.395 | 2.393 | 2.372 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 688   | 775   | 811   | 821   | 844   | 848   | 822   | 804   | 797   | 737   | 650   | 631   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 62    | 83    | 108   | 128   | 127   | 124   | 128   | 118   | 97    | 78    | 60    | 52    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.70  | 1.39  | 2.44  | 4.05  | 4.94  | 5.24  | 4.96  | 4.33  | 3.12  | 1.76  | 0.81  | 0.51  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.10  | 0.25  | 0.32  | 0.56  | 0.57  | 0.42  | 0.61  | 0.41  | 0.41  | 0.24  | 0.13  | 0.06  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |  |  |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|--|--|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |  |  |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |  |  |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | +0.53 | +0.60 | N/A         | N/A     | N/A     | N/A     |  |  |
| (47) Regional (36 neighbors) |  | +0.40 | N/A     | N/A    | N/A     | N/A   | +0.33 | N/A         | N/A     | +52     | N/A     |  |  |

Nomenclature: See separate page